

PCB Routing Notes

ECET-230 Air Mouse Project

PCB Layout Revision Notes

Revision 1

- Imported schematic into KiCad PCB Editor
- Placed ATmega32U4 microcontroller near center of board
- Added MPU6050 sensor placement for short I2C routing
- Positioned USB Micro connector at edge of PCB for accessibility

Revision 2

- Adjusted component spacing to reduce routing congestion
- Added vias to route traces between layers
- Reworked D+ and D- USB traces for cleaner layout
- Added pull-up resistors for SDA and SCL communication lines

Revision 3

- Added ground plane to improve grounding and reduce noise
- Routed 3.3V power traces across PCB
- Optimized trace paths around microcontroller pins
- Reduced overlapping air wires by moving traces to additional copper layers

Final PCB Adjustments

- Verified all required nets were connected
- Checked GND and power routing continuity
- Updated silkscreen labels for readability
- Saved final PCB screenshots and exported project files

Design Challenges

- Limited PCB space around the ATmega32U4 pins
- Managing USB routing without crossing traces
- Routing power traces while maintaining organization
- Balancing trace spacing and via placement

Final Notes

The PCB was completed using KiCad 9.0 with multi-layer routing techniques. Vias and internal routing layers were used to simplify trace management and improve overall board organization. The final layout includes USB communication, motion sensor integration, power regulation, and user input buttons for the air mouse prototype.